전산화학 (2018년 1학기) 2차 시험 해답

(총 **7문제, 120점**)

1. **(총 20점, 각 10점)**

(a) $f(x,y) = x^2 + y^2 + 2x$ 가 (x^*, y^*) 에서 극값을 가지므로

$$f_x(x^*, y^*) = 0 \Rightarrow 2x^* + 2 = 0 \Rightarrow x^* = -1$$
$$f_y(x^*, y^*) = 0 \Rightarrow 2y^* = 0 \Rightarrow y^* = 0$$

그러므로 **임계점은 (-1,0)** 이다. Hessian 검사를 이용하면

$$\Delta_1 = f_{xx}(x^*, y^*) \quad \text{그리고} \quad \Delta_2 = f_{xx}(x^*, y^*) f_{yy}(x^*, y^*) - f_{xy}(x^*, y^*)^2.$$

주어진 함수 $f(x,y) = x^2 + y^2 + 2x$ 에 대하여,

$$f_{xx}(x,y) = 2$$
$$f_{yy}(x,y) = 2$$
$$f_{xy}(x,y) = 0$$

그러므로

$$\Delta_1 = 2 > 0 \quad \text{and} \quad \Delta_2 = 2 \times 2 - 0^2 = 4 > 0$$

$\Delta_1 > 0$ 이고, $\Delta_2 > 0$ 이므로 $f(x,y)$ 는 $(-1,0)$ 에서 극소값을 갖는다.

(b) $f(x,y)$ 의 임계점을 (x^*, y^*) 라 하면, 임계점에서

$$f_x(x^*, y^*) = 0 \Rightarrow 3(x^*)^2 - 3 = 0 \Rightarrow x^* = \pm 1$$
$$f_y(x^*, y^*) = 0 \Rightarrow 2y^* - 4 = 0 \Rightarrow y^* = 2$$

그러므로 **임계점은 (1,2), (-1,2)** 이다. Hessian 검사를 이용하면

$$\Delta_1 = f_{xx}(x^*, y^*) \quad \text{그리고} \quad \Delta_2 = f_{xx}(x^*, y^*) f_{yy}(x^*, y^*) - f_{xy}(x^*, y^*)^2.$$

주어진 함수 $f(x,y) = x^3 + y^2 - 3x - 4y + 2$ 에 대하여

$$f_{xx}(x,y) = 6x$$
$$f_{yy}(x,y) = 2$$
$$f_{xy}(x,y) = 0$$

i. $(1,2)$ 에서

$$\begin{aligned} f_{xx}(1,2) &= 6 \\ f_{yy}(1,2) &= 2 \\ f_{xy}(1,2) &= 0 \end{aligned}$$

그러므로

$$\Delta_1 = 6 > 0 \quad \text{그리고} \quad \Delta_2 = (6)\times(2) - (0)^2 = 12 > 0$$

$\Delta_1 > 0$ 이고, $\Delta_2 > 0$ 이므로 $f(x,y)$ 는 $(1,2)$ 에서 극솟값을 갖는다.

ii. $(-1,2)$ 에서

$$\begin{aligned} f_{xx}(-1,2) &= -6 \\ f_{yy}(-1,2) &= 2 \\ f_{xy}(-1,2) &= 0 \end{aligned}$$

그러므로

$$\Delta_1 = -6 < 0 \quad \text{그리고} \quad \Delta_2 = (-6)\times(2) - (0)^2 = -12 < 0$$

$\Delta_2 < 0$ 이므로 $f(x,y)$ 는 $(-1,2)$ 에서 안장점을 갖는다.

2. **(10점)**

$$\begin{aligned} \left(\frac{\partial f(x,y)}{\partial x}\right)_y &= \cos x\cos y + \frac{1}{y} \\ \left(\frac{\partial}{\partial y}\left(\frac{\partial f(x,y)}{\partial x}\right)_y\right)_x &= -\cos x\sin y - \frac{1}{y^2} \qquad (1) \\ \left(\frac{\partial f(x,y)}{\partial y}\right)_x &= -\sin x\sin y - \frac{x}{y^2} \\ \left(\frac{\partial}{\partial x}\left(\frac{\partial f(x,y)}{\partial y}\right)_x\right)_y &= -\cos x\sin y - \frac{1}{y^2} \qquad (2) \end{aligned}$$

(1) 식과 (2) 식을 비교해보면 같음을 알 수 있다. 그러므로

$$\left(\frac{\partial}{\partial y}\left(\frac{\partial f(x,y)}{\partial x}\right)_y\right)_x = \left(\frac{\partial}{\partial x}\left(\frac{\partial f(x,y)}{\partial y}\right)_x\right)_y$$

0이므로 Euler 상관관계식0이 성립한다.

3. **(총 20점, 각 5점)**

(a) $df(x,y) = (4x+3y)dx + (3x+8y)dy$

$$M(x,y) = 4x+3y, \qquad N(x,y) = 3x+8y$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = 3$$
$$N_x(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = 3$$
$$M_y(x,y) = N_x(x,y)$$

완전미분, $f(x,y) = 2x^2 + 3xy + 4y^2 + C$

(b) $df(x,y) = y\cos x dx + \sin x dy$

$$M(x,y) = y\cos x, \qquad N(x,y) = \sin x$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = \cos x$$
$$N_y(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = \cos x$$
$$M_y(x,y) = N_x(x,y)$$

완전미분, $f(x,y) = y\sin x + C$

(c) $df(x,y) = (x^2-y)dx + (y^2+x)dy$

$$M(x,y) = (x^2-y), \qquad N(x,y) = (y^2+x)$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = -1$$
$$N_x(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = 1$$
$$M_y(x,y) \neq N_x(x,y)$$

불완전미분.

(d) $df(T,V) = C_V(T)dT + \frac{RT}{V}dV$

$$M(T,V) = C_V(T), \qquad N(x,y) = \frac{RT}{V}$$
$$M_V(T,V) = \left(\frac{\partial M}{\partial V}\right)_T = 0$$
$$N_T(T,V) = \left(\frac{\partial N}{\partial T}\right)_V = \frac{R}{V}$$
$$M_V(T,V) \neq N_T(T,V)$$

불완전 미분.

4. **(10 점)**

$$
\begin{aligned}
w &= -\int_{\bar{V}_1}^{\bar{V}_2} P d\bar{V} \\
&= -\int_{\bar{V}_1}^{\bar{V}_2} \left(\frac{RT}{\bar{V}-B} - \frac{A}{T^{1/2}\bar{V}(\bar{V}+B)} \right) d\bar{V} \\
&= -\int_{\bar{V}_1}^{\bar{V}_2} \left\{ \frac{RT}{\bar{V}-B} - \frac{A}{BT^{1/2}} \left(\frac{1}{\bar{V}} - \frac{1}{\bar{V}+B} \right) \right\} d\bar{V} \\
&= -\left[RT\ln(\bar{V}-B) - \frac{A}{BT^{1/2}} \Big(\ln \bar{V} - \ln(\bar{V}+B) \Big) \right]_{\bar{V}_1}^{\bar{V}_2} \\
&= -\left\{ RT\ln(\bar{V}_2-B) - \frac{A}{BT^{1/2}} \Big(\ln \bar{V}_2 - \ln(\bar{V}_2+B) \Big) \right\} + \left\{ RT\ln(\bar{V}_1-B) - \frac{A}{BT^{1/2}} \Big(\ln \bar{V}_1 - \ln(\bar{V}_1+B) \Big) \right\} \\
&= -RT\ln\left(\frac{\bar{V}_2-B}{\bar{V}_1-B} \right) - \frac{A}{BT^{1/2}} \ln\left(\frac{\bar{V}_1(\bar{V}_2+B)}{\bar{V}_2(\bar{V}_1+B)} \right)
\end{aligned}
$$

5. **(10 점)** $f(x,y,z) = 2x^3 - 3x(y^2+z^2)$

$$
\begin{aligned}
\nabla^2 f(x,y,z) &= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \\
&= 12x - 6x - 6x = 0
\end{aligned}
$$

6. **(총 20점, 각 10점)**

(a)

$$
\begin{aligned}
&\int_0^{2\pi} d\phi \int_0^{\pi} d\theta Y_0^0(\theta,\theta) Y_2^0(\theta,\theta) \sin\theta \\
&= \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \left(\frac{1}{4\pi} \right)^{1/2} \left(\frac{5}{16\pi} \right)^{1/2} (3\cos^2\theta - 1) \sin\theta \\
&= \left(\frac{1}{4\pi} \right)^{1/2} \left(\frac{5}{16\pi} \right)^{1/2} \int_0^{2\pi} d\phi \int_0^{\pi} (3\cos^2\theta - 1) \sin\theta d\theta \\
&= \left(\frac{1}{4\pi} \right)^{1/2} \left(\frac{5}{16\pi} \right)^{1/2} (2\pi) \int_0^{\pi} (3\cos^2\theta - 1) \sin\theta d\theta
\end{aligned}
$$

$t = \cos\theta$ 라고 치환하면, $dt = -\sin\theta d\theta$

$$
\begin{aligned}
&= -\left(\frac{1}{4\pi} \right)^{1/2} \left(\frac{5}{16\pi} \right)^{1/2} (2\pi) \int_1^{-1} (3t^2 - 1) dt \\
&= \left(\frac{1}{4\pi} \right)^{1/2} \left(\frac{5}{16\pi} \right)^{1/2} (2\pi) \int_{-1}^{1} (3t^2 - 1) dt \\
&= \left(\frac{1}{4\pi} \right)^{1/2} \left(\frac{5}{16\pi} \right)^{1/2} (2\pi) \Big[t^3 - t \Big]_{-1}^{1} \\
&= 0
\end{aligned}
$$

(b)

$$\int_0^{2\pi} d\phi \int_0^{\pi} d\theta Y_1^0(\theta,\phi)(\cos\theta)Y_2^0(\theta,\phi)\sin\theta$$
$$= \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \left(\frac{1}{4\pi}\right)^{1/2}(\cos\theta)\left(\frac{5}{16\pi}\right)^{1/2}(3\cos^2\theta-1)\sin\theta$$
$$= \left(\frac{1}{4\pi}\right)^{1/2}\left(\frac{5}{16\pi}\right)^{1/2}\int_0^{2\pi} d\phi \int_0^{\pi}\cos\theta(3\cos^2\theta-1)\sin\theta d\theta$$
$$= \left(\frac{1}{4\pi}\right)^{1/2}\left(\frac{5}{16\pi}\right)^{1/2}(2\pi)\int_0^{\pi}\cos\theta(3\cos^2\theta-1)\sin\theta d\theta$$

$t=\cos\theta$ 라고 치환하면, $dt=-\sin\theta d\theta$

$$= -\left(\frac{1}{4\pi}\right)^{1/2}\left(\frac{5}{16\pi}\right)^{1/2}(2\pi)\int_1^{-1} t(3t^2-1)dt$$
$$= \left(\frac{1}{4\pi}\right)^{1/2}\left(\frac{5}{16\pi}\right)^{1/2}(2\pi)\int_{-1}^{1}(3t^3-t)dt$$

$3t^3-t$ 는 기함수이므로

$$= 0$$

∴ **전이불가능하다.**

7. **(총 30점)**

(a) **(10점)**

$$\int_C \Big[xydx+(x^2-y^2)dy\Big], \qquad (C:(0,0)\to(1,2)\ \text{직선 경로})$$

직선 경로이므로 $y=2x$

$$= \int_0^1 x(2x)dx + \int_0^2 \left[\left(\frac{y}{2}\right)^2 - y^2\right]dy$$
$$= \int_0^1 2x^2dx - \frac{3}{4}\int_0^2 y^2dy$$
$$= \left[\frac{2}{3}x^3\right]_0^1 - \frac{3}{4}\left[\frac{1}{3}y^3\right]_0^2 = \frac{2}{3}-2 = -\frac{4}{3}$$

(b) **(10점)**

$$\int_C (x^2ydx+xy^2dy), \qquad (C:(0,0)\to(1,0)\to(1,2))$$
$$= \int_0^1 x(0)dx + \int_0^2 (1^2-y^2)dy$$
$$= \left[y-\frac{1}{3}y^3\right]_0^2 = -\frac{2}{3}$$

(c) **(5점)** (a)와 (b)의 결과에서 경로에 따라 선적분 값이 달라지므로 완전미분이 아니다.

(d) **(5점)** $M(x,y) = xy,\ N(x,y) = x^2 - y^2$ 로 놓으면,

$$\left(\frac{\partial M}{\partial y}\right)_x = x, \qquad \left(\frac{\partial N}{\partial x}\right)_y = 2x \Rightarrow \left(\frac{\partial M}{\partial y}\right)_x \neq \left(\frac{\partial N}{\partial x}\right)_y$$

이므로, $x^2ydx + xy^2dy$ 는 불완전미분이다.